

KACHUG, Russian FederationWMO: **306220**Lat: **53.967N**Lon: **105.900E**Elev: **1709**StdP: **13.81**Time Zone: **8.00 (E08)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.4	-40.8	-49.4	0.3	-44.6	-45.2	0.4	-40.3	13.1	5.5	11.0	4.2	0.2	160	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	85.2	65.7	81.6	64.0	78.3	62.8	68.5	79.7	66.6	76.9	64.8	74.7	4.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.8	97.9	73.4	63.0	91.8	71.5	61.0	85.3	69.8	34.0	79.4	32.3	77.1	30.9	74.9	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	10.8	9.1	-47.2	91.6	6.4	3.1	-51.8	93.8	-55.5	95.6	-59.1	97.4	-63.7	99.7		
(5)			-46.5	72.8	5.6	3.4	-50.5	75.2	-53.8	77.2	-57.0	79.1	-61.0	81.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	26.6	-17.8	-9.6	11.6	34.0	46.9	60.0	65.1	59.6	45.7	29.3	4.9	-13.1		
	DBStd	30.84	13.85	13.51	13.13	8.06	7.71	6.73	4.99	6.07	7.28	9.26	14.63	14.34		
	HDD50	9734	2102	1668	1192	481	153	7	0	7	171	643	1354	1956		
	HDD65	14134	2567	2088	1657	930	562	174	61	183	580	1107	1804	2421		
	CDD50	1181	0	0	0	1	57	308	468	305	41	1	0	0		
	CDD65	106	0	0	0	0	1	25	64	16	0	0	0	0		
	CDH74	1812	0	0	0	0	89	552	836	314	21	0	0	0		
	CDH80	468	0	0	0	0	16	144	240	66	2	0	0	0		
	WSAvg	2.9	1.0	1.4	2.7	4.5	5.0	4.0	3.4	3.2	3.2	3.0	1.9	1.2		
	PrecAvg	12.2	0.3	0.2	0.2	0.4	0.9	1.9	2.6	3.0	1.5	0.5	0.4	0.4		
(15) Precipitation	PrecMax	17.5	0.6	0.5	0.6	0.9	1.9	4.5	6.9	6.2	3.4	1.3	0.8	1.0		
	PrecMin	8.7	0.0	0.0	0.0	0.0	0.2	0.5	0.7	1.1	0.4	0.0	0.1	0.2		
	PrecStd	2.2	0.1	0.1	0.1	0.2	0.5	1.0	1.5	1.3	0.8	0.3	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.7	28.8	45.7	67.0	82.2	89.0	89.9	86.3	76.9	60.0	38.5	23.9		
		MCWB	20.3	25.8	36.7	48.6	59.0	64.5	69.1	66.2	59.4	48.1	33.2	21.7		
	2%	DB	14.3	21.9	40.1	59.9	76.0	84.0	85.9	81.1	70.1	52.4	32.9	19.5		
		MCWB	13.1	19.5	32.9	45.0	55.7	63.5	67.4	64.2	56.2	43.0	29.9	18.3		
	5%	DB	8.0	16.1	35.6	54.2	70.8	80.3	82.4	77.4	64.2	47.1	29.2	13.6		
		MCWB	7.0	14.2	30.3	41.7	52.6	62.0	65.9	63.0	53.3	40.3	27.0	12.5		
	10%	DB	2.4	10.6	31.7	49.1	65.1	76.6	78.7	73.5	59.2	42.6	25.2	8.2		
		MCWB	1.5	9.3	27.6	39.0	49.9	60.9	64.7	61.6	51.4	37.2	23.4	7.3		
	0.4%	WB	20.5	25.8	37.3	49.6	60.6	69.1	72.8	69.5	61.2	48.8	33.6	22.0		
		MCDB	22.0	29.1	44.6	63.4	78.6	81.6	85.0	80.4	72.6	58.2	37.9	23.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	13.1	19.6	33.3	45.8	56.7	66.3	69.8	66.4	57.8	44.2	30.1	18.3		
		MCDB	14.5	22.1	39.2	58.3	73.4	78.0	81.7	77.4	68.1	51.1	32.6	19.5		
	5%	WB	7.0	14.1	30.5	42.6	53.6	64.2	67.8	64.4	54.7	40.8	27.0	12.7		
		MCDB	7.9	16.0	35.3	52.9	68.6	76.6	78.0	74.3	62.9	46.3	29.2	13.7		
	10%	WB	1.6	9.2	27.6	39.4	50.7	62.2	66.1	62.5	51.8	37.6	23.4	7.3		
		MCDB	2.4	10.6	31.7	48.5	63.2	73.8	75.6	71.8	58.4	42.5	25.1	8.1		
	5% DB	MDBR	16.1	22.0	26.7	22.9	26.6	26.1	23.2	23.5	22.7	20.1	17.7	15.5		
		MCDBR	19.7	25.4	26.5	32.8	39.0	34.4	32.1	31.8	33.6	28.8	19.2	18.9		
		MCWBR	18.7	22.9	19.9	19.3	20.6	15.9	14.9	16.1	19.2	19.5	16.3	18.0		
		MCWBR	19.8	25.3	25.0	30.8	35.6	28.7	25.8	27.6	29.6	27.8	19.0	18.9		
(40) Clear-Sky Solar Irradiance	taub		0.273	0.302	0.329	0.370	0.414	0.458	0.447	0.391	0.343	0.318	0.294	0.272		
		taud	2.050	2.030	2.058	2.200	2.154	2.103	2.183	2.331	2.421	2.375	2.169	2.032		
	Ebn at Noon		207	242	265	271	264	252	252	262	263	242	200	178		
		Edh at Noon	23	33	40	40	44	47	43	35	28	23	20	19		
(44) All-Sky Solar Radiation	RadAvg		349	691	1189	1564	1710	1841	1689	1455	1015	615	338	241		
	RadStd		20	25	47	110	126	110	145	67	86	60	24	19		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	-3.31	N/A	N/A	-0.94	N/A	N/A	+79	N/A	
(47) Regional (10 neighbors)		N/A	N/A	-4.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page